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Submitted by: Viraj Patel Sankhla (W2155089)

Predicting S&P 500 Stock Market Volatility: The Role of Macroeconomic Variables and the VIX

Abstract

Using macroeconomic variables, this study examines the predictability of S&P 500 monthly realised volatility from May 2001 to December 2025 (211 observations). The evidence demonstrates that lagged VIX is the primary volatility predictor using OLS regression with Newey-West standard errors, a GARCH-X(1,1) model, a Vector Autoregression, and Granger causality tests. All five macro variables Granger-cause realised volatility, though the lagged VIX alone accounts for the bulk of in-sample fit. A straightforward autoregressive benchmark performs better out-of-sample than more complex macro specifications, indicating the possibility of overfitting in volatile regimes.

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1. Introduction

The volatility of the stock market is the cornerstone of financial risk management. The ability to forecast market volatility has theoretical and practical ramifications for everything from portfolio allocation to option pricing. The S&P 500 is a natural subject since it is the most widely used benchmark for international stocks. Its volatility has been both chronic and episodic during times of crisis, such as the COVID-19 shock of 2020 and the Global Financial Crisis (GFC) of 2008-2009, clustering silently for years before rising sharply.

Whether macroeconomic fundamentals provide information regarding equities volatility beyond what historical volatility itself reveals is a recurrent subject in the empirical finance research. After Schwert (1989) demonstrated the relationship between macro variables and stock market volatility, Engle, Ghysels, and Sohn (2013) formalised this relationship using the GARCH-MIDAS framework, which enables low-frequency economic data to be incorporated into high-frequency volatility models. More recently, Paye (2012) expressed doubts regarding macroeconomic predictors' out-of-sample utility, pointing out that true forecast improvements are not always correlated with statistical in-sample significance.

This study poses a specific question: Can monthly realised volatility of the S&P 500 be predicted using lagged macroeconomic data, including the CBOE VIX, unemployment rate, CPI inflation, industrial production growth, and the 10-year Treasury yield? Three frameworks for econometrics are used. First, a linear baseline is produced via an OLS regression using Newey-West standard errors. Second, whether these variables enhance conditional variance modelling is tested using a GARCH-X(1,1) model. Third, the dynamic relationship between volatility and the macroenvironment is captured via system that includes impulse response functions and Granger causality tests.

Combining these complementary techniques on a lengthy post-2000 sample that covers several volatility regimes makes a significant contribution. The findings present a complex picture: in-sample macro factors are dominated by VIX. Granger-cause realised volatility, but out-of-sample, a straightforward autoregressive model outperforms all macro formulations.

2. Literature Review and Theoretical Framework

The theoretical basis for macro-volatility linkages is the notion that uncertainty about business cash flows and discount rates, which both contribute to equity risk, is a direct result of uncertainty about economic fundamentals. According to Merton's (1973) intertemporal CAPM, state factors that describe the investment opportunity set exactly the kind of variables assessed by yields, inflation, and unemployment and should affect risk premia and, consequently, volatility.

The spline-GARCH, which breaks down conditional variance into a smooth long-run component driven by macroeconomic fundamentals and a short-run GARCH component, was first presented by Engle and Rangel (2008). Their data on global indices demonstrated that the long-term volatility pattern is significantly driven by GDP growth and inflation. Using MIDAS regressions to combine monthly macro data with daily returns in the GARCH-MIDAS model, Engle, Ghysels, and Sohn (2013) expanded on this and discovered that inflation and growth in industrial production aid in predicting S&P 500 volatility.

The VIX should be treated differently. The VIX is a natural predictor of realised volatility since it represents the implied volatility of S&P 500 options, which encodes forward-looking market expectations of volatility. When predicting short-horizon volatility, Blair, Poon, and Taylor (2001) showed that VIX incorporates nearly all information from past returns. Because the VIX consistently surges during crises, Whaley (2000) described it as a "fear gauge," a term that has endured.

Paye (2012) looked at a variety of macro predictors for the out-of-sample question and discovered that, even though they appeared significant in-sample, most were unable to outperform a basic random walk or AR benchmark when forecasting realised variance out-of-sample. This issue is pertinent in this situation. Corsi's (2009) HAR model demonstrated that lagged realised volatility is a strong predictor over a variety of time periods, offering a natural standard by which more complex models must be judged.

Theoretically, the business cycle and the unemployment rate are related because rising unemployment indicates economic slack, which increases earnings uncertainty and may lead to more widespread risk aversion. Interest rates move through a variety of pathways; depending on the situation, a rising 10-year yield may indicate tighter financial conditions or better growth

prospects, with opposing volatility consequences. While industrial production acts as a proxy for real activity, CPI inflation increases the uncertainty around monetary policy response.

When considered collectively, the evidence lends credence to a conditional view: macro factors are important, particularly during regime transitions, but their marginal predictive value beyond implied volatility and past realised volatility may be constrained during stable times. This is the main conflict that is examined in this examination.

3. Data Description and Exploration

211 full observations from the Bloomberg Terminal are included in the collection, which spans monthly observations from May 2001 to December 2025. Monthly realised volatility was calculated as the annualised standard deviation of daily log returns within each calendar month, and monthly log returns were constructed using the price of the S&P 500 index. The implied volatility measure was derived from the CBOE VIX index. US unemployment (%), CPI year-over-year inflation (%), industrial production year-over-year growth (%), and the 10-year US Treasury yield (%) were examples of macroeconomic variables. To ensure consistency, all series were downloaded monthly.

Several characteristics of the data are shown by the summary statistics. The distribution shows negative skewness (-1.12) and excess kurtosis of 2.46, as supported by the Jarque-Bera test statistic of 96.85 ($p < 0.001$), which rejects normality. Monthly log returns averaged 0.81% with a standard deviation of 5.06%. This is in line with stylised realities in equity returns that have been thoroughly studied. Realised volatility ranged from 0.055 to 0.326, with the COVID-19 disaster at the high end and an average of 0.161. Throughout the sample, the VIX averaged 19.74, which is close to its traditional stress threshold of 20. The 10-year yield averaged 3.16%, the CPI averaged 2.61%, and the unemployment rate averaged 5.72%.

Variable	Mean	Std Dev	Min	Max
Monthly Log Return	0.0081	0.0506	-0.228	0.127
Realised Volatility	0.161	0.067	0.056	0.326
VIX Level	19.74	9.62	11.4	59.9
Unemployment (%)	5.72	2.19	3.4	14.7
CPI YoY (%)	2.61	2.09	-2.1	9.1
Ind. Production YoY (%)	0.61	4.93	-15.1	14.0
10Y Treasury Yield (%)	3.16	1.42	0.51	5.25

Table 1: Summary Statistics (May 2001 to December 2025, $N = 211$). Source: Bloomberg Terminal.

Monthly returns ($p < 0.01$) and industrial production growth ($p < 0.01$) are stationary, according to Augmented Dickey-Fuller (ADF) tests, while log realised volatility, log VIX, unemployment, CPI, and the 10-year yield do not reject the unit root null at the 5% level. Given their known use in practical forecasting work, a common pragmatic decision when near-unit-root behaviour rather

than a true unit root is likely all five non-stationary variables were employed in levels in the VAR. The adoption of GARCH-family models was justified by an ARCH-LM test on the return series ($\text{Chi-sq} = 13.85$, $p = 0.031$), which verified the existence of ARCH effects.

Figure 1 depicts the S&P 500 price level rising sharply from about 1,100 at the beginning of the sample to over 6,500 by the end of 2025, with notable declines during the GFC (2008-09), COVID-19 (2020), and the inflation shock of 2022. With the worst single-month return of roughly -23% in March 2020, the return series (Figure 2) demonstrates volatility clustering. Realised volatility (Figure 3) was much higher than the sample mean of 0.161, peaking at 0.33 during the GFC and surpassing 0.34 during COVID-19.

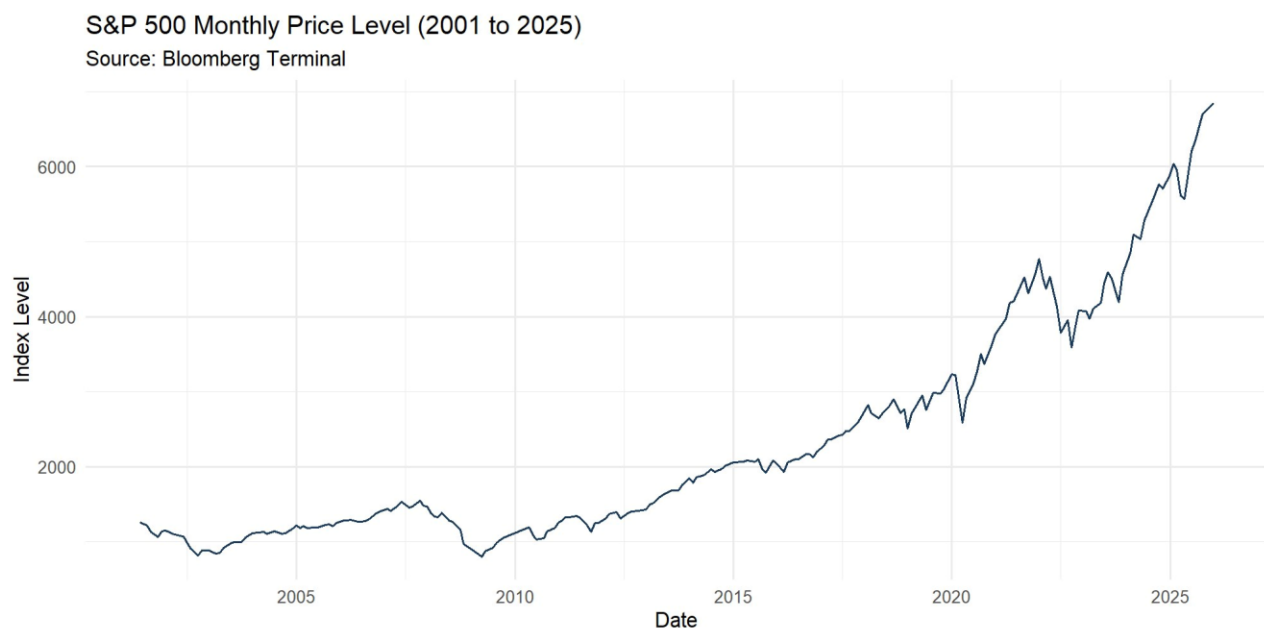


Figure 1: S&P 500 Monthly Price Level (2001-2025). Source: Bloomberg Terminal.

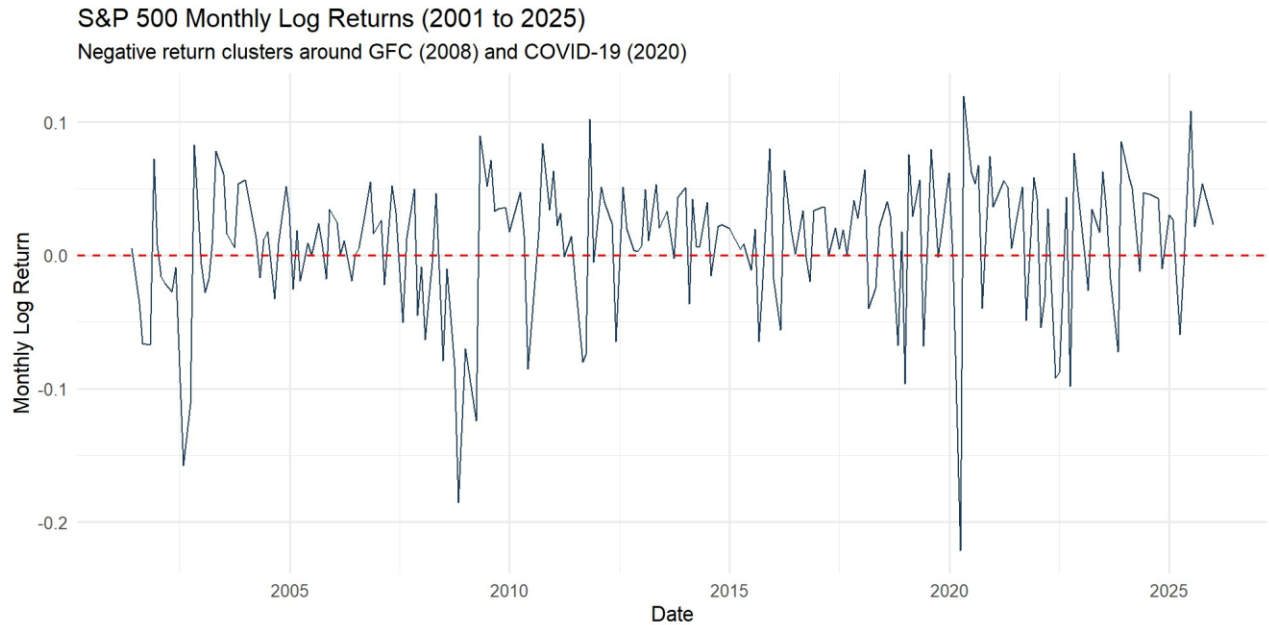


Figure 2: S&P 500 Monthly Log Returns (2001-2025).

Negative clusters visible at GFC (2008) and COVID-19 (2020).

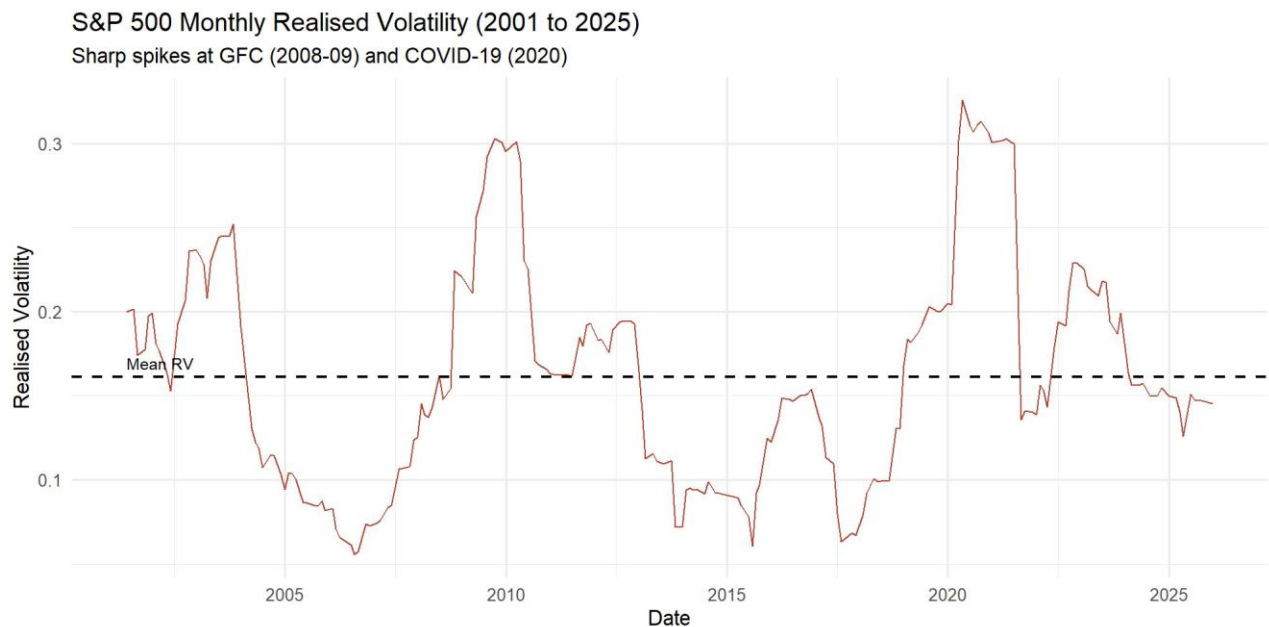


Figure 3: S&P 500 Monthly Realised Volatility (2001-2025).

Dashed line marks the sample mean of 0.161.

A graphic explanation for the VIX series' function as a fear proxy can be seen in Figure 4 in the appendix, which surged to almost 60 during the GFC and above 52 during COVID-19. VIX is supported as the main predictor in the OLS formulation by the scatter plot of log VIX against log

realised volatility, which displays a significant positive linear connection. The macroeconomic predictors across time are depicted in Figure 5, where the two most notable structural alterations of the sample period are the CPI jump above 9% in mid-2022 and the unemployment spike to about 15% in April 2020.

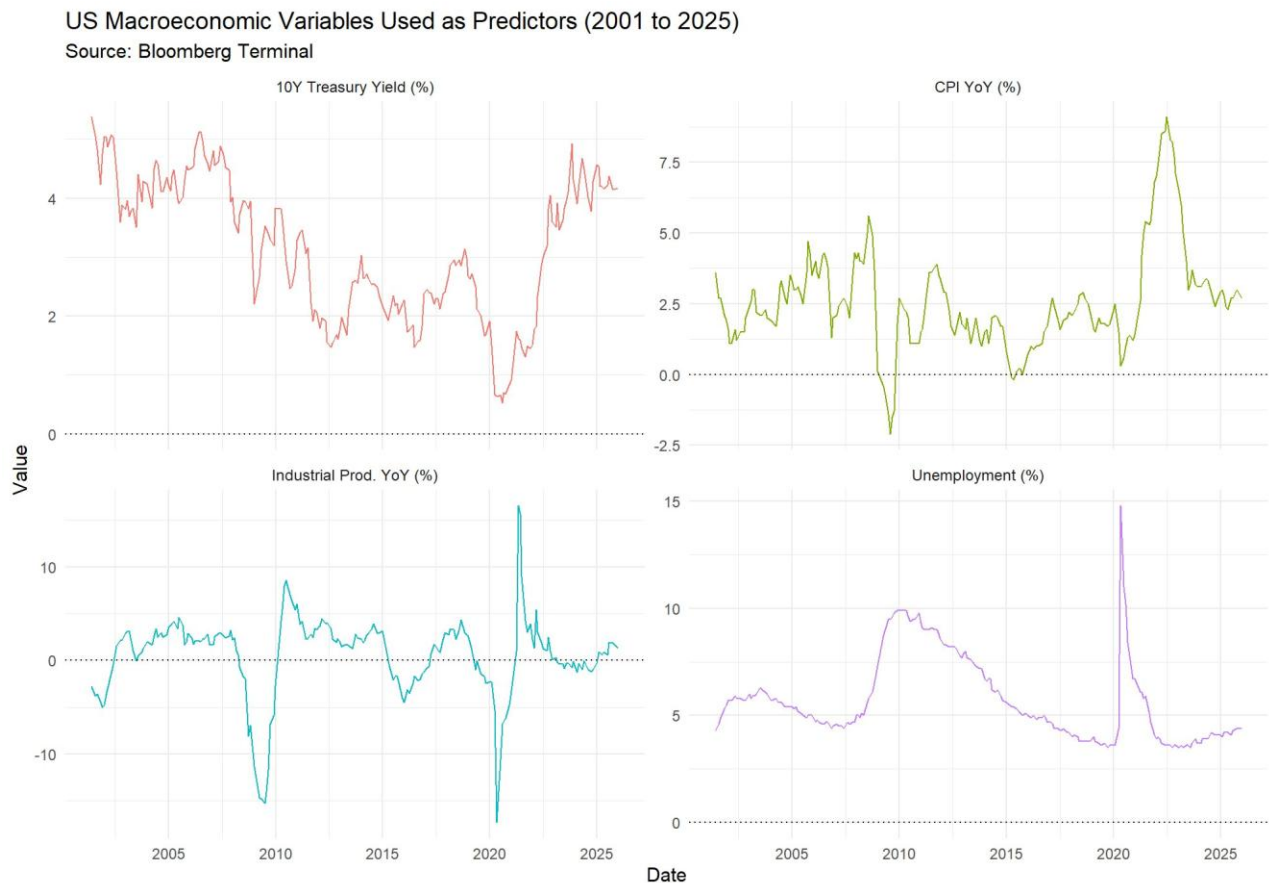


Figure 4: US Macroeconomic Predictors (2001-2025): 10-Year Yield, CPI YoY, Industrial Production YoY, and Unemployment. Source: Bloomberg Terminal.

4. Econometric and Statistical Modelling Approach

We estimate three complimentary models. Each focus on a distinct facet of the macro-volatility relationship: system-level causality, non-linear conditional variance dynamics, and linear predictability.

4.1 OLS Baseline with Newey-West Standard Errors

The primary specification regresses log realised volatility on one-month-lagged macro predictors to preserve a genuine forecasting structure and avoid any look-ahead bias:

$$\log(RV_t) = \alpha + \beta_1 \log(VIX_{t-1}) + \beta_2 UNEMP_{t-1} + \beta_3 CPI_{t-1} + \beta_4 INDPROD_{t-1} + \beta_5 YIELD_{t-1} + \varepsilon_t$$

To increase linearity and lessen the impact of extreme values, log transforms are performed to both realised volatility and VIX. In time-series regressions of this kind, residual serial correlation is typically adjusted by applying Newey-West heteroskedasticity and autocorrelation consistent (HAC) standard errors with three lags.

4.2 GARCH-X(1,1)

A GARCH-X(1,1) model is estimated with the five lagged macro variables entering the variance equation as external regressors to reflect the well-known time-varying and clustering nature of equity volatility. For the conditional variance, the model uses the conventional GARCH(1,1) form, enhanced as:

$$\sigma^2_t = \omega + \alpha \varepsilon^2_{t-1} + \beta \sigma^2_{t-1} + \gamma' X_{t-1}$$

where γ represents the lag macro regressors' contribution to the conditional variance level and X_{t-1} is their vector. The innovation distribution is represented as Student-t to account for the fat tails shown in the data, and the mean equation contains an AR(1) factor. Model selection is based on the Akaike and Bayesian information criteria, and a conventional GARCH(1,1) without external regressors is estimated as a comparator.

4.3 Vector Autoregression and Granger Causality

All five macro variables and log realised volatility are used to estimate a VAR system, with lag length $p = 2$ chosen by AIC. When conditioning on its own lags, Granger causality tests determine whether each variable has statistically significant predictive content for log realised volatility. Impulse response functions (IRFs) use 95% bootstrap confidence bands built from 100 replications over a 12-month period to trace the dynamic response of log realised volatility to a one-standard-deviation shock in each variable.

5. Results and Interpretation

5.1 OLS Baseline Results

The OLS model provides a reasonably good in-sample fit for a monthly macro forecasting regression, explaining around 48% of the variation in log realised volatility ($R^2 = 0.4816$, adjusted $R^2 = 0.4689$). Each of the five regressors is statistically significant under traditional standard errors. However, only log VIX (lag 1) maintains significance at conventional levels (coefficient = 0.634, $t = 4.26$, $p < 0.001$) after using Newey-West corrections to account for the serial correlation in the residuals. Following the HAC correction, the remaining macro variables: unemployment, industrial production, CPI, and the 10-year yield lose relevance.

Variable	Coefficient	HAC t-stat	p-value
Intercept	-3.851	-7.565	< 0.001***
Log VIX (lag 1)	0.634	4.262	< 0.001***
Unemployment (lag 1)	0.031	1.008	0.315
CPI YoY (lag 1)	0.033	1.076	0.283
Ind. Production (lag 1)	-0.021	-1.644	0.102
10Y Yield (lag 1)	-0.053	-1.011	0.313

Table 2: OLS Regression Results with Newey-West (3-lag) Standard Errors.

Dependent variable: $\log(\text{Realised Volatility})$. $N = 210$. $R^2 = 0.482$.

A 10% increase in the VIX is linked to around a 6.3% increase in realised volatility the next month, according to the log VIX coefficient of 0.634, which is significant from an economic standpoint. Although this is not statistically significant after the HAC correction, the sign on industrial production is negative (-0.021), which is consistent with the notion that higher real activity is linked to reduced market stress. The idea that higher rates occasionally indicate a more benign macro backdrop, at least at the monthly horizon, is also consistent with the negative yield coefficient (-0.053).

5.2 GARCH-X(1,1) Results

With a persistence sum of 0.815 and well-defined ARCH and GARCH parameters of $\alpha_1 = 0.334$ and $\beta_1 = 0.481$, the GARCH-X(1,1) estimation showed significant but not near-unit-root volatility persistence. The fat-tailed Student-t distribution fits the data well, as indicated by the shape parameter of 4.98. Nevertheless, the variance equation's five external macro regressors (vxreg1 through vxreg5) all produced coefficients that were essentially at zero, with t-statistics near zero and p-values more than 0.99. This suggests that the lagged macro variables do not contribute any new information to the conditional variance equation once the ARCH and GARCH dynamics are taken into consideration.

This finding is supported by comparing information criteria. Despite having five extra parameters, the GARCH-X model produced a worse AIC of -3.2153 and BIC of -3.0400 than the normal GARCH(1,1), which achieved an AIC of -3.2629 and BIC of -3.1672. The sequential nature of GARCH already absorbs the information contained in the macro regressors, which penalise model complexity without boosting fit. Both models pass the diagnostic tests with ease; Ljung-Box tests on squared residuals and standardised residuals reveal no ARCH effects or residual serial correlation.

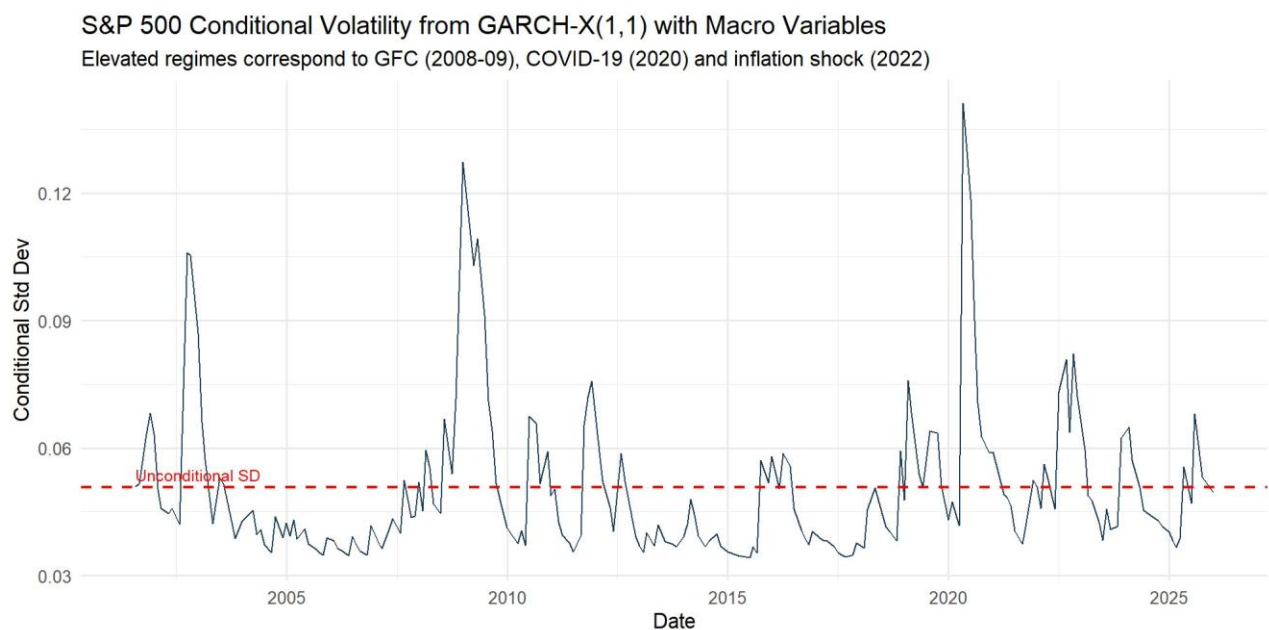


Figure 5: Conditional Volatility from GARCH-X(1,1).

The red dashed line marks the unconditional standard deviation of returns (approx. 5.06%).

5.3 Granger Causality and Impulse Responses

A more intriguing picture is revealed by the VAR Granger causality tests. At the 5% level, all five macro variables: log VIX ($F = 4.44$, $p < 0.001$), unemployment ($F = 3.07$, $p < 0.001$), CPI ($F = 1.95$, $p = 0.035$), industrial production ($F = 3.01$, $p < 0.001$), and the 10-year yield ($F = 2.14$, $p = 0.019$). This is a more robust result than the OLS experiment suggests; each variable has predictive content for realised volatility inside a system that conditions on its own history.

Variable	F-statistic	p-value	Significance
Log VIX	4.445	< 0.001	***
Unemployment	3.072	0.0007	***
CPI YoY	1.955	0.0349	**
Industrial Production	3.008	0.0009	***
10Y Treasury Yield	2.140	0.0192	**

Table 3: Granger Causality Tests. H_0 : Variable does not Granger-cause log realised volatility. VAR(2) with constant.

The direction and duration of these impacts can be inferred from the impulse response functions. Realised volatility increases monotonically after a one-standard-deviation shock to the VIX, peaking between months five and six at about 0.068 log units before progressively declining over the next twelve months (Figure 6). Within the 95% confidence zones, the total response stays well above zero, indicating a strong and long-lasting influence. In contrast, the unemployment shock results in a negative response (Figure 7): realised volatility initially decreases after an unemployment shock. This is counterintuitive, but it could be due to the VAR system's lagged response or the confounding of unemployment with post-crisis stabilisation phases when volatility was already declining.

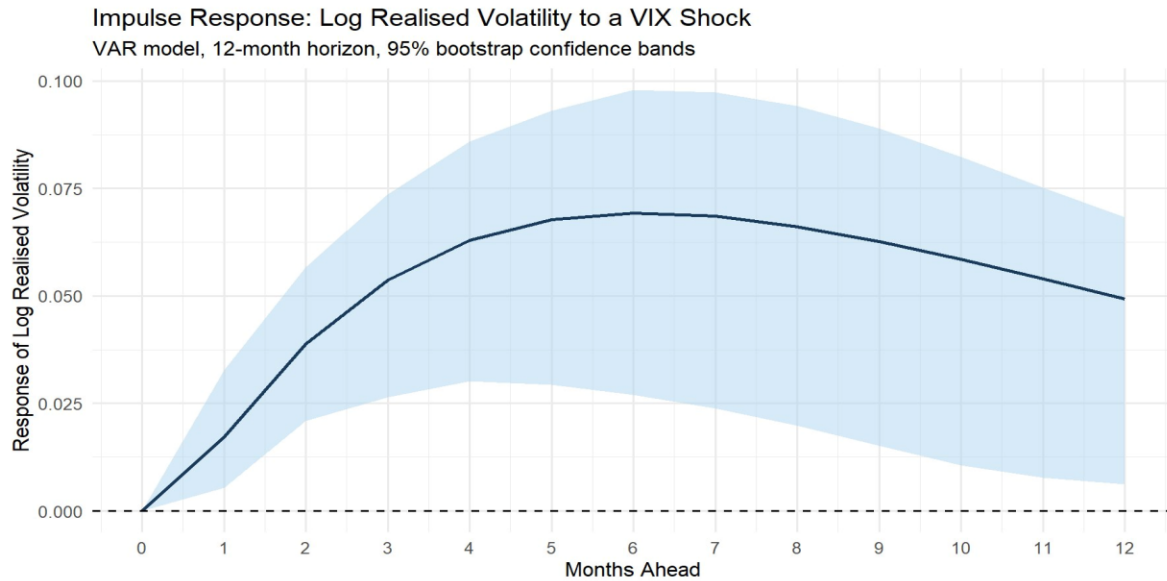


Figure 6: Impulse Response of Log Realised Volatility to a VIX Shock. VAR(2), 12-month horizon, 95% bootstrap confidence bands.



Figure 7: Impulse Response of Log Realised Volatility to an Unemployment Shock.

The confidence bands include zero throughout, indicating weak statistical precision.

5.4 Out-of-Sample Forecast Evaluation

A hold-out sample was set aside for the last 24 months of data (2024 to 2025). Three models were assessed: an OLS model using lagged VIX only, an OLS model with all five macro variables, and a straightforward AR benchmark using only lagged log realised volatility. The outcomes are remarkable.

Model	Out-of-Sample RMSE
AR Benchmark (lagged log RV)	0.06716
OLS with VIX only	0.33200
OLS with all macro variables	0.40577

Table 4: Out-of-Sample Forecast RMSE. Hold-out period: final 24 observations (approx. 2024-2025).

Both macro requirements are significantly outperformed by the AR benchmark. This is in line with Paye (2012) and the more general conclusion drawn from the research on variance forecasting: historical realised volatility is a challenging standard to surpass. Despite having an in-sample fit of about 48%, the macro-OLS models yield subpar out-of-sample predictions, with RMSE values four to six times higher than the benchmark. This is probably due to parameter instability and the potential for macro-volatility linkages to change across regimes, especially during the 2022-2023 inflation phase and the post-COVID recovery.

6. Conclusion and Future Research

The purpose of this study was to determine whether macroeconomic factors might forecast the monthly realised volatility of the S&P 500 over a 24-year sample that included several crises. Although inconsistent, the evidence is consistent. Granger causality tests verify that all five macro variables have predictive capacity within a VAR framework, and the lagged VIX explains the majority of volatility variation in-sample with a R^2 close to 48%. A VIX shock causes a positive, long-lasting, and statistically distinct impulsive reaction. These results are consistent with the larger body of research on macro-volatility linkages and Engle et al. (2013).

However, the GARCH-X results reveal that once past squared shocks and conditional variance are controlled for, the macro variables contribute nothing to the conditional variance equation. Their information appears to be already absorbed by the GARCH dynamics. More critically, the out-of-sample exercise demonstrates that the macro-based OLS models are handily beaten by a naive AR benchmark, with RMSE ratios of roughly 5:1. This supports the cautionary findings of Paye (2012) and suggests that the in-sample significance of macro predictors does not reliably translate into genuine forecast improvements.

The fact that all models are estimated on the entire sample and presume parameter stability across drastically diverse types of crises is one of the study's main limitations. In actuality, the correlation between, for example, unemployment and volatility during the Great Financial Crisis may differ significantly from that during the epidemic. The next logical step would be to use rolling or recursive estimating algorithms. Additionally, the monthly frequency might be too coarse to capture the fast-moving nature of volatility during crisis periods; predictive power that monthly OLS misses could be recovered by switching to weekly data or combining daily and monthly information using the GARCH-MIDAS technique.

Non-linear threshold specifications, in which macro variables only affect volatility when they surpass stress levels, such as unemployment above 7% or the VIX above 30, may also be investigated in future studies. The linear benchmarks presented could be used to test machine learning techniques like random forests and LASSO, which have also demonstrated potential in macro forecasting applications. Lastly, it would be possible to determine whether the findings are unique to the US market after 2000 or apply the same paradigm to other key equity indices by expanding the sample backward to cover the 1990s.

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Appendix

A. CBOE VIX and VIX-RV Scatter

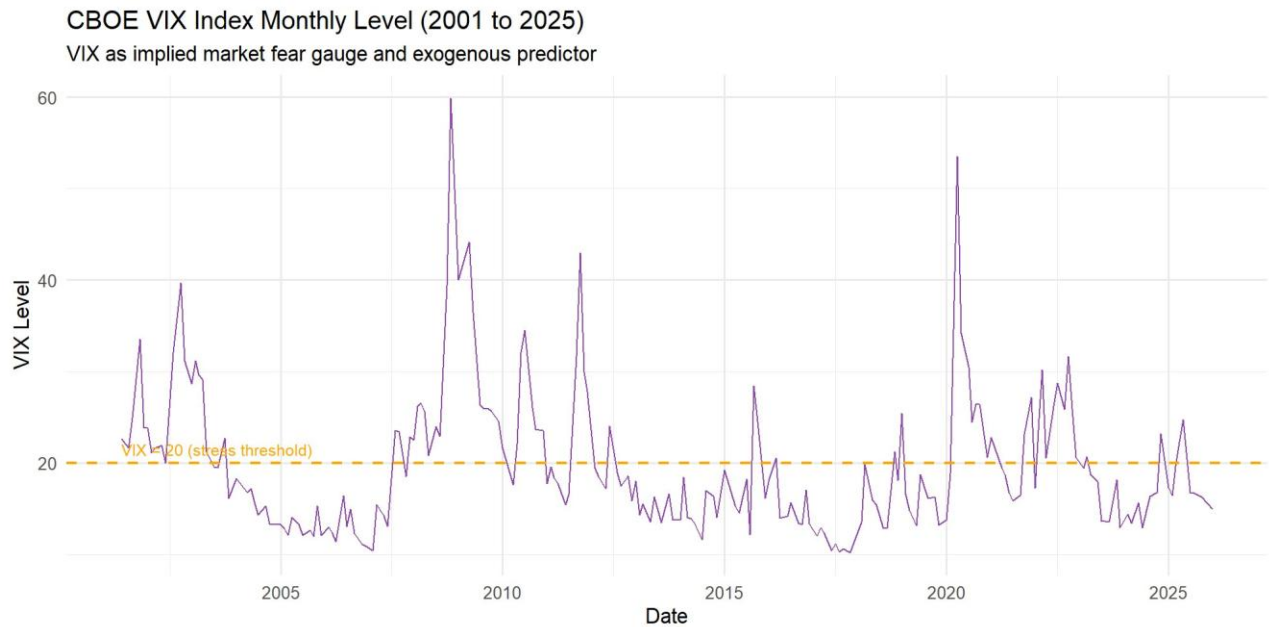


Figure A1: CBOE VIX Index Monthly Level (2001-2025). Orange dashed line marks the conventional stress threshold of $VIX = 20$.

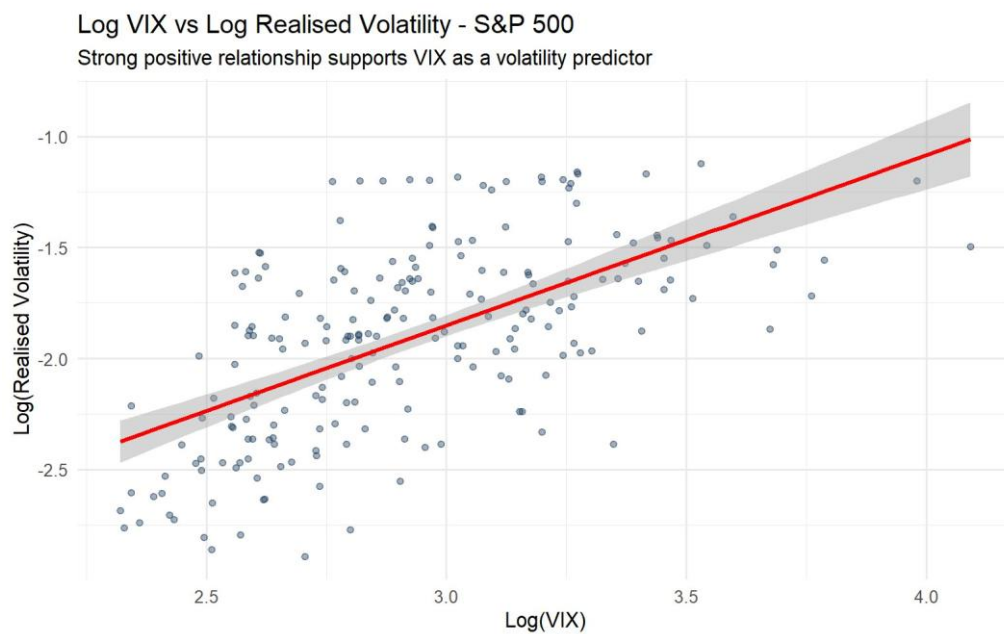


Figure A2: Log VIX vs Log Realised Volatility. The strong positive relationship ($R^2 \approx 0.48$) justifies VIX as the primary predictor.

B. ADF Unit Root Test Results

Variable	ADF Statistic	p-value	Decision
Returns	-5.440	< 0.01	Stationary
Log Realised Volatility	-3.019	0.149	Non-stationary
Log VIX	-2.984	0.164	Non-stationary
Unemployment	-2.370	0.421	Non-stationary
CPI YoY	-3.432	0.051	Non-stationary
Industrial Production	-5.147	< 0.01	Stationary
10Y Yield	-1.516	0.779	Non-stationary

Table A1: Augmented Dickey-Fuller Unit Root Tests. H_0 : Series has a unit root.

C. GARCH(1,1) vs GARCH-X(1,1) Information Criteria

Model	AIC	BIC	Log-Likelihood
GARCH(1,1) Baseline	-3.2629	-3.1672	348.600
GARCH-X(1,1) with Macro	-3.2153	-3.0400	348.606

Table A2: Information Criteria Comparison. Lower AIC/BIC indicates better fit adjusted for model complexity.